

SCHEDULE-ALIGNED IDENTITY-LOCK PERIODICITY UNDER AN ACTIVE-BLOCK RECORDER

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ABSTRACT

Position-decomposed diagnostics of diffusion large language models can confuse sampler geometry with model behavior because an active-block recorder exposes positions only through the current sampling window. We hold the model, questions, canvas, decoding rule, and total function evaluations fixed while applying a composite intervention: block partition changes together with the required per-block step allocation. Across five tested arms, the registered candidate selector and a non-extremal rank-sum comparison both select the configured block length. This is selection-and-stability evidence, not a null-calibrated detection result: the L24 margin is marginal ($p_{\text{margin}} = 7.1 \times 10^{-2}$), and the L48 separation rests on one boundary locus. Separately registered non-nested probes on a canvas-96 bed do not reveal a recorder-observable fixed-32 component. We recommend reporting separately what boundary-set composition makes reachable before data and what measured magnitudes support. The composite intervention, reused L32 development control, post-treatment scope amendment, and untested cross-model, cross-schedule, and cross-recorder scope preclude a model-mechanism claim.

1 INTRODUCTION

Diffusion language models generate discrete sequences by iterative denoising (Austin et al., 2021; Zheng et al., 2024; Lou et al., 2024; Shi et al., 2024; Nie et al., 2025). As decoding schedules become variable or structured (Li et al., 2025; Sun et al., 2026; Wang et al., 2026), a position-resolved diagnostic can inherit periodicity from the schedule that exposes its positions. Without an intervention, such a readout cannot distinguish model-intrinsic state, sampler state, and what the recorder makes observable.

We examine identity-lock time on a blockwise-decoded LLaDA-style model (Nie et al., 2025; 2026). An apparent period of 32 motivates a matched composite intervention over the block partition and its required per-block step allocation: the model, 300 questions, canvas, total NFE, decoding rule, and temperature remain fixed. The active-block recorder, however, logs only masked positions in the currently active block. Its lock-time profile is therefore defined relative to the sampling window (Figure 1).

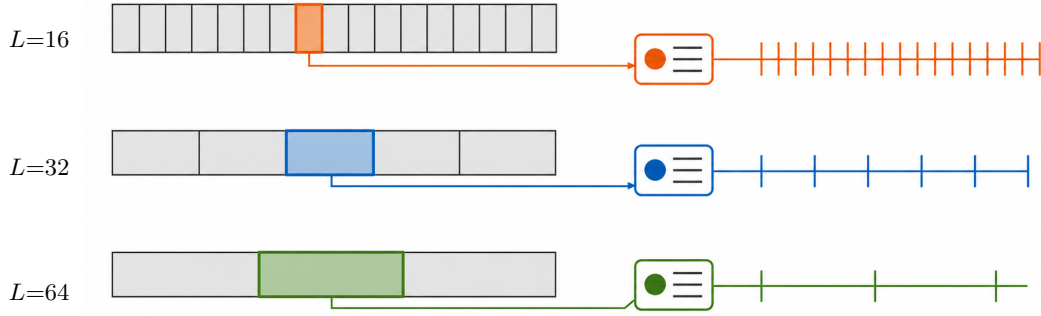


Figure 1: Qualitative measurement pipeline. From top to bottom, block length L grows while canvas extent stays fixed; color marks the one active block exposed to the recorder, and the right-hand ticks depict the boundary spacing visible through that window. Pulse and group counts are schematic, not quantitative evidence. Measured profiles and separations are in Figure 2 and Table 3.

Our evidence makes three scoped contributions. First, across five tested arms, the registered candidate selector and a non-extremal rank-sum comparison both return $P^* = L$; this is a descriptive selection-and-stability result rather than a null-calibrated period test. Second, the measured candidate-32 signs and separately registered non-power-of-two probes do not reveal a fixed-32 component at the recorder-observable layer; they do not rule out an unobserved model or sampler component. Third, this case study motivates a reporting recommendation: specify before data which outcomes boundary-set composition makes reachable, then report separately what the measured magnitudes show.

The evidential limitations are part of those contributions. The L24 corrected margin is marginal at $p_{\text{margin}} = 7.1 \times 10^{-2}$, the L48 estimate rests on one boundary locus, and the reused L32/NFE48 arm is a development control. Moreover, the originally registered model-intrinsic branch was withdrawn after treatment (amendment v3; Appendix A). The paper therefore audits a recorder-relative measurement on the tested beds; it does not establish a causal subtype or a mechanism across models, schedules, or recorders.

2 MEASUREMENT AND DESIGN

Core notation is defined here before it is used; Appendix C retains the longer terminology crosswalk.

Notation. For decode i and relative canvas position r , $\ell_{i,r}$ is the recorded identity-lock time. Its positionwise median is $m(r)$ and $d(r) = m(r+1) - m(r)$. For candidate period P , $\mathcal{B}_P$ contains candidate boundary indices and $\mathcal{I}_P$ their interior complement; $\text{sep}(P)$ is the boundary-minimum minus the interior-maximum of $d(r)$, and P^* maximizes it. “Canvas 96” denotes the separate 96-position probe bed. BIN_k denotes a restricted content null with k equally populated offset-rank strata.

Model bed. We use an LLaDA-style masked dLLM at the LLaDA-8B-Instruct scale with a locally registered checkpoint on GSM8K (Cobbe et al., 2021). Decoding uses temperature 0 and `remasking=low_confidence`, the confidence-ordered parallel decoding rule introduced for conditional masked language models and used by LLaDA (Ghazvininejad et al., 2019; Nie et al., 2025). The main differential fixes NFE at 48 and uses the same 300 questions. Its L32/NFE48 arm is a frozen *development control*: the registered contiguous `decode_id` $\in [345, 645)$ subset of a pre-existing pool. The L32/NFE32 content bed is separate and supports only Section A.7.

Setup and protocol. Table 1 consolidates every bed and schedule. The headline length metric is $d(r) = m(r+1) - m(r)$, the adjacent difference of median identity-lock time. For content, the same lock-time ordering is compared with residualized confidence under an explicit eight-stratum offset-rank null (BIN8; four-position granularity), plus a same-amplitude *position-fake*: confidence values

Element	Measured arms	L32 control	Non-nested probes	Content bed
Canvas / NFE	256/48	256/48	96/48	256/32
L ; per-block steps	16 : $[4] \times 8 + [2] \times 8$; 64 : $[16, 16, 8, 8]$	32 : $[8] \times 4 + [4] \times 4$	24 : $[12] \times 4$; 48 : $[24] \times 2$	32
Questions	same 300	contiguous [345, 645), $n=300$	same 300	GSM8K
Candidate grid	{8, 16, 32, 64}	{8, 16, 32, 64}	{8, 16, 24, 32, 48, 64}	—
Resampling	period $B=2000$; stability/sign $B=1000$	period $B=2000$	period $B=2000$	interval $B=500$; null $R=2000$

Table 1: Unified configuration. All beds use the same model family, task, temperature, and remasking rule. The control’s contiguous [345, 645) subset is not part of the non-nested probes. Those two arms form a separate canvas-96 setup with $n=300$, NFE 48, the displayed schedules and grid, and 2,000 cluster-bootstrap replicates. NFE denotes total function evaluations. At fixed total NFE, changing the partition also changes the required per-block step allocation; the design is therefore a composite intervention and does not isolate an allocation effect.

are assigned to positions to preserve amplitude while destroying the observed content-to-position pairing. Intervals use cluster-bootstrap percentiles over units (Davison & Hinkley, 1997); decode resampling preserves multiplicity.

Identity-lock time. For decode i , relative canvas position r , and recorded step s , let $x_{i,r,s}$ be the predicted token and $x_{i,r}^f$ its final value. We define

$$\ell_{i,r} = \min\{s : x_{i,r,t} = x_{i,r}^f \text{ for every recorded } t \geq s\}. \quad (1)$$

Thus, $\ell_{i,r}$ is the earliest recorded step after which the token identity never changes again. The position profile is $m(r) = \text{median}_i \ell_{i,r}$ and its adjacent difference is $d(r) = m(r+1) - m(r)$. All analysis uses relative generated-canvas position, not prompt position.

Min-max period separation and registered selection criterion. For each candidate period in the registered grid (Table 3), define the candidate boundary set $\mathcal{B}_P = \{r : r \bmod P = P-1\}$ and interior $\mathcal{I}_P$ as its complement. The registered estimator is

$$\text{sep}(P) = \min_{r \in \mathcal{B}_P} d(r) - \max_{r \in \mathcal{I}_P} d(r), \quad P^* = \arg \max_P \text{sep}(P). \quad (2)$$

Exact ties select the first (smallest) period in the ordered registered grid. Positive separation requires every candidate boundary jump to exceed every interior jump. On a strictly L -supported profile it penalizes divisors with false boundaries and multiples that leave true boundaries inside; without that condition it is not a general divisor/multiple guarantee. The registered selection-and-stability criterion fires only under the conjunction $\text{sep}(P^*) > 0$ and a cluster-bootstrap modal period equal to P^* with mode fraction at least 0.95. This operational criterion is not a null-calibrated hypothesis test: its bootstrap mode fraction measures stability under decode resampling, not a false-positive rate. The same conjunction classifies each bootstrap readout below. We resample the 300 decodes with replacement, preserving multiplicity, for $B=2,000$ replicates.

Selection stability and its honest reading of “no misses.” Two resampling readouts are reported. First, a *joint paired cluster bootstrap* resamples a shared index set of the 300 decodes once and applies the same indices to all five arms (the matched three plus the two canvas-96 probes), so the headline is tested as a *cross-arm relation* on shared questions rather than as five independent marginals. Across $B=2,000$ replicates (seed 20260824), $\Pr[P_{16}^*=16 \wedge P_{32}^*=32 \wedge P_{64}^*=64] = 1.0000$; each of all five arms also has marginal $\Pr[P^*=L] = 1.0$ (Table 9). Second, we report the probability that the registered criterion actually fires, $\Pr[\text{sep}(L) > 0]$, instead of the “0 misses” language. Table 2 consolidates this readout across all five arms. Because the lock times are integer (temperature 0), the min-max separation is an integer-valued extremum of integer medians, so the bootstrap lower bound can sit exactly at the point estimate and “0 misses” reflects discrete selection stability rather than an absence of error. We therefore do not read a 0/2,000 miss count as false-positive control.

Because sep is an extremum statistic, we also report a *non-extremal* robustified comparison: a rank-sum (Mann–Whitney U) separation between candidate-boundary and candidate-interior values of $d(r)$. Its argmax period equals L on *all five* arms, including the L32 control where the min–max separation was weakest; the per-arm statistics are in Table 2 and their provenance in Table 9. A quantile-based separation survives only on L64 and L48 and selects multiples otherwise, which we report honestly as partial rather than as a full vote for the extremum statistic.

Replication accounting. Table 1 is the complete resampling inventory. Period selection and its interval share one draw, whereas the sign and stability checks use separate draws; Table 3 identifies the shared readouts. Uncounted values are observed-data point estimates; artifacts are in Table 9.

Matched intervention and scope. The two treatment arms and frozen development control are in Table 1. At fixed total NFE, changing the partition necessarily changes per-block step allocation, so the treatment is partition-plus-required-allocation. A within-arm rank transform preserves $P^*=L$ and each candidate-32 sign on all five tested block-length arms, including the non-power-of-two $L=24$ and $L=48$ (Table 9); a uniform-allocation arm was not registered. The result therefore identifies a recorder-output association with the composite partition-plus-allocation schedule, not which component places or scales the boundary. A factorial allocation/partition intervention is future work.

Registration and two-layer reachability. The estimator and selection-and-stability criterion were fixed before treatment, but amendment v3 withdrew the model-intrinsic decision branch after treatment; the timestamped chronology is in Appendix A, Table 8. The canvas-96 probes were registered separately before their execution. We therefore report reachability at two layers. *A priori*, nesting such as $\mathcal{B}_{32} \subset \mathcal{B}_{16}$ determines set composition but not the sign of $\text{sep}(32)$. *From events*, the observed magnitude ordering gives $\text{sep}(32) = -2$ on L16 and a negative value on L64. Thus a recorder-observable fixed-32 component is not selected under these active-block regimes; this does not identify model-intrinsic or sampler-state causes. The exact loci and synthetic condition are consolidated in Table 6; the legacy verdict labels remain only in Appendix C.

Separately conditioned content analysis. On the separate L32/NFE32 bed, BIN8 permutes confidence within eight equally-populated offset-rank bins after a fixed nuisance fit. The initial confidence conf_0 is read at the first recorded step of a position’s block, following confidence-ordered parallel decoding (Ghazvininejad et al., 2019; Nie et al., 2025). This supporting diagnostic is not a cross-block-length validation.

3 RESULTS

3.1 THE REGISTERED SELECTOR FOLLOWS BLOCK LENGTH

The development control recovers its known period, and both treatment arms move to their respective block lengths (Figure 2; Table 3). Only the selected period has positive min–max separation, and

Readout	L32		L24		L48
	L16	control	L64	probe	probe
$\Pr[\text{sep}(L) > 0]$, joint bootstrap	1.0	0.8805	0.9995	0.999	1.0
Rank-sum z at L	13.9	8.3	4.4	3.9	2.0
Wrong-candidate fire rate, original three-bed probe	0.0	0.0	0.0	—	—

Table 2: Selection stability and non-extremal robustness (not a false-positive rate). The first row uses one $B=2,000$ joint paired bootstrap and reports how often the conjunction $\text{sep}(L) > 0$ and mode fraction ≥ 0.95 fires. The L32 control’s 0.8805 is the weakest arm and appears alongside the perfect modal-argmax relation. The second row is the Mann–Whitney comparison, whose maximizing candidate is L on every arm. The final row is a separate $B=1,000$ in-place wrong-candidate diagnostic for the original three beds: its zeros follow from negative point estimates and are not fresh-null false-positive rates. Within an arm, cells share draws. Provenance is in Table 9.

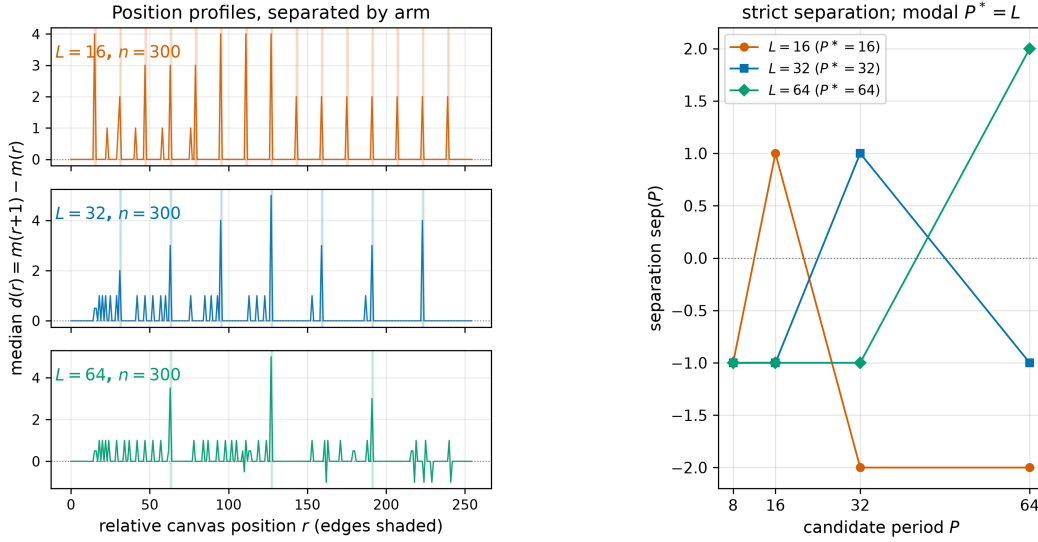


Figure 2: Data-derived differential result. Left: three stacked facets isolate the median adjacent-difference profile $d(r) = m(r+1) - m(r)$ for L16 (orange), the L32 frozen control (blue), and L64 (green), each from $n=300$ decodes; shaded ticks mark that arm’s block edges. The control is the contiguous $\text{decode_id} \in [345, 645)$ subset. Right: only the matching candidate has positive separation in each arm. Regenerated by `scripts/r265_fig_dprofile_overlay.py`; quantitative values come from the registered events, and the recorder-level comparison does not distinguish sampler from model mechanism.

the bootstrap selects it in every replicate. The paired cross-arm relation holds in all 2,000 shared-cluster resamples, while a rank-sum point estimator independently selects $P^*=L$ in all five arms. Across the five arms, P^* leads the best wrong candidate by +2.0 to +8.0 d -units; four reach 5% significance against the specific within-block jitter null (exact $p_{\text{margin}} \leq 2.1 \times 10^{-2}$), while L24 does not, with $p_{\text{margin}} = 7.1 \times 10^{-2}$ (Table 5). Thus four arms reject this particular null and L24 does not. This red-team comparison is neither an arm-level false-positive-rate estimate nor family-wise calibration of the complete candidate-selection procedure.

Candidate 32 under the active-block recorder. Table 3 reports $\text{sep}(32) = -2$ on the measured $L = 16$ arm. The exact values in Table 6 show why: candidate-32 boundary values remain positive, but the measured odd-16 interior is larger. The joint sign $\text{sep}(16) > 0 \wedge \text{sep}(32) < 0$ holds in all 1,000 dedicated bootstrap replicates. Candidate 32 is therefore not selected at the recorder-observable layer on this bed. Nesting alone does not force that sign, and the result says nothing about an unobserved model-intrinsic or sampler-state component.

Probe arm	Non-nested $\text{sep}(32)$	Disclosed marginal candidate
$L=24$ probe	-8.0	$P=48$: -1.0; band $[-3.0, 0.0]$
$L=48$ probe	-7.5	—

Table 4: Separate canvas-96 non-nested probes. Candidate 32 is disjoint from the $L=24$ boundaries. The L24 marginal cell concerns the mismatched $P=48$ candidate rather than selected $P^*=24$; the displayed percentile band is a selection-stability diagnostic, not a coverage-valid interval.

L (source)	$\text{sep}(P)$				P^* (mode frac.)	$\text{sep}(P^*)$ stability band [†]	$(\min_B, \max_I)$		
	$P = 8$	$P = 16$	$P = 32$	$P = 64$			@ P^*	$ B_{P^*} $	$ I_{P^*} $
16 (measured)	-1	+1	-2	-2	16 (1.0)	[+1, +1]	(+2.0, +1.0)	15	240
32 (control)	-1	-1	+1	-1	32 (1.0)	[0, +2]	(+2.0, +1.0)	7	248
64 (measured)	-1	-1	-1	+2	64 (1.0)	[+1, +3]	(+3.0, +1.0)	3	252
24 (probe)	—	—	—	—	24 (1.0)	[+1.5, +4.0]	(+5.0, +1.0)	3	92
48 (probe)	—	—	—	—	48 (1.0)	[+4.0, +6.5]	(+8.0, +1.5)	1	94

Table 3: Development-control, differential, and probe results. Each arm uses 2,000 cluster-bootstrap replicates; the mode fraction and percentile band are paired readouts of the same draw. The $(\min_B, \max_I)$ columns report the boundary minimum and interior maximum, followed by their observed set cardinalities. Every arm selects its configured block length. Because lock times are integer-valued, the bands diagnose selection stability rather than coverage: L16 collapses to a point mass, while the L48 probe rests on one boundary location. Tables 1, 2, 5, 7, and 11 give the complete resampling inventory and separately budgeted checks; provenance is in Table 9.

Arm	P^*	$\text{sep}(P^*)$	m_{obs}	p_{sepL}	p_{margin}
L16 (canvas 256)	16	+1.0	+2.0	5.0×10^{-4}	1.5×10^{-3}
L64 (canvas 256)	64	+2.0	+3.0	5.0×10^{-4}	1.0×10^{-3}
L24 (canvas 96)	24	+4.0	+5.0	5.0×10^{-4}	7.1×10^{-2}
L48 (canvas 96)	48	+6.5	+8.0	2.3×10^{-2}	2.1×10^{-2}
L32 control (canvas 256, NFE 48)	32	+1.0	+2.0	5.0×10^{-4}	1.0×10^{-3}

Table 5: Separation-margin selectivity calibration. Each of $R=2,000$ replicates (seed 20260825) permutes $d(r)$ within each true L -block, including its edge index, then records the argmax candidate’s top-minus-second gap. Here $m_{\text{obs}} = \text{sep}(P^*) - \max_{c \neq P^*} \text{sep}(c)$ and the discrete exact $p = (1 + \#\{m_{\text{perm}} \geq m_{\text{obs}}\}) / (R + 1)$. The p_{sepL} column tests $\text{sep}(P^*)$ alone; p_{margin} is the selection-controlled readout. Provenance and the original-gate size audit are in Table 9 and Appendix A.5.

Measured class	Loci	$d(r)$ values	Readout
L32 true-32 boundaries	[31, 63, 95, 127, 159, 191, 223]	[2, 3, 4, 5, 3, 3, 4]	development control
L16 true-16 odd chain	[15, 47, 79, 111, 143, 175, 207, 239]	[4, 3, 3, 4, 2, 2, 2, 2]	candidate-32 interior max 4
L16 would-be $\mathcal{B}_{32}$	[31, 63, 95, 127, 159, 191, 223]	[2, 3, 4, 4, 2, 2, 2]	boundary min 2; sep(32) = -2
L64 mid-block 32	[31, 95, 159, 223]	L64: [0, 0, 0, 0]; L32: [2, 4, 3, 4]	recorder-observable collapse
L64 true-64	[63, 127, 191]	L64: [3.5, 5, 3]; L32: [3, 5, 3]	true boundary remains

Strict-support synthetic L	Source periods	Selected P^*
16	{8, 16, 32, 48}	16 for every source
32	{8, 16, 32}	32 for every source
64	{8, 16, 32}	64 for every source

Table 6: Consolidated negative evidence, reachability, and development check. A priori, nested boundary sets determine membership but do not force the sign. From events, L16 candidate 32 has boundary minimum 2 and interior maximum 4; the joint sign holds in 1,000/1,000 bootstrap replicates. On L64, would-be 32 boundaries are recorder-observable interiors. The lower panel is a development-only synthetic control: $P^* = L$ holds for the shown strictly L -supported profiles, not for arbitrary residuals. The upper rows are observed-data readouts from their arm samples, not additional replications. Provenance is in Table 9.

3.2 SELECTION STABILITY, NOT NULL CALIBRATION OR MECHANISM

The central estimand is discrete period identity. On the observed samples, positive min-max separation is a deterministic property of the recorded profiles; the bootstrap then probes the stability of period selection to resampling decodes. The selected period is stable under both the per-arm bootstrap and the joint paired bootstrap that re-applies one shared question-index draw to all five arms (Section 2); the registered criterion fires in 0.88–1.0 of resamples across arms, with the L32 development control the weakest at $\Pr[\text{sep}(32) > 0] = 0.8805$. A robustified non-extremal rank-sum separation still selects L on all five arms. This does not establish an arbitrarily small effect or a universal period: it establishes stable selection among the registered candidates on the stated bed, and the low-factor cells ($|B_{48}|=1$ on the L48 probe) temper any margin reading. The L32 control also reuses a pre-existing contiguous [345, 645] pool and its percentile stability band $[0, +2]$ touches zero (it is not a coverage-valid CI). These three weaknesses are disclosed together; a fresh in-session L32 rerun is GPU-gated future work.

Exploratory spectral residual check. Because sep cannot exclude every residual shape for a nested candidate, a post-data DFT check tests one sinusoidal residual-32 family on L16. Its injected-residual control fires and its no-residual control is silent; the real L16 profile does not cross the family-wise threshold at $P = 32$, and a direct boundary-subset test finds no amplitude split. This is a narrow negative for the tested family, not evidence that arbitrary residual shapes are absent. The estimator, complete candidate grid, power sweep, and thresholds are reported in Appendix A.1 and Table 7.

Canvas-96 non-nested probe (boundary disclosure). On the separate canvas-96 probe beds, candidate 32 is *non-nested* ($\mathcal{B}_{32} \cap \mathcal{B}_{24} = \emptyset$), and the same estimator does not select a recorder-observable fixed-32 component without relying on the canvas-256 nesting relation. Table 3 reports the two probe estimates, their selected block lengths, and the disclosed marginal mismatched cell (Table 4); the probes used about 0.38 GPU-h. This is a scoped extension (canvas 96 $\neq$ 256), not a replacement for the matched canvas-256 arms.

The L64 mid-block reversal and its L32 comparator are reported without a separate verdict vocabulary in the consolidated negative-evidence Table 6.

Supporting analysis outside the headline claim. A separately budgeted L32/NFE32 bed shows a moderate-granularity association between initial confidence and later lock ordering. Because it is not a cross-block-length validation and does not identify the five-arm mechanism, its estimand, results, and null-granularity audit are reported in Appendix A.7.

4 LIMITATIONS

- **Observation is not mechanism.** The active-block recorder cannot separate model-intrinsic state, sampler state, and recorder-observable state. Candidate 32 is not selected only at the last layer under the tested regimes.
- **Registration history.** The estimator was selected with a failed development control before treatment, but the model-intrinsic decision branch was withdrawn after treatment. The non-nested probes have their own pre-run registration and a different 96-position canvas.
- **Bed and grid.** Evidence covers one model family, one task, and the candidate grids in Table 1; off-grid periods, other models, tasks, samplers, and untested block lengths are not covered.
- **Composite intervention.** Partition and required per-block step allocation vary together. Boundary location is measured; a distinct allocation effect on amplitude is not.
- **Boundary cardinality.** The min-max separation is an extremum statistic, so its support matters: on the probe arms the selected period rests on few measured boundary positions (the L48 probe has $|B_{48}|=1$), making its positive separation a single-locus readout rather than a stable margin. We report $|B_{P^*}|$ and $|I_{P^*}|$ per arm in Table 3 for this reason.
- **Recorder and control arms.** A dual-recorder comparison that logs the same trajectory through active-block and full-canvas/passive instruments is GPU-gated future work; no cross-recorder result is claimed. A fresh in-session L32/NFE48 run is also GPU-gated: the present control reuses a pre-existing contiguous pool, has the lowest positivity (0.8805), and its percentile stability band touches zero.
- **Content scope.** The content association comes from one L32/NFE32 bed and survives only through moderate null granularity (it is not established at the finest registered BIN32 granularity). Its behavior on the L16/L64 beds is not measured.

5 RELATED WORK

Foundational discrete-diffusion work develops structured denoising, reparameterization, density-ratio estimation, and simplified masked diffusion (Austin et al., 2021; Zheng et al., 2024; Lou et al., 2024; Shi et al., 2024). These studies concern the model or sampler trajectory rather than an active-block recorder perturbed at fixed model and NFE.

The nearest schedule-axis methods include fully parallel and semi-autoregressive decoding (Gu et al., 2018; Wang et al., 2018), confidence-ordered mask-predict and LLaDA-style decoding (Ghazvininejad et al., 2019; Nie et al., 2025; 2026), and variable or reversible schedules (Li et al., 2025; Sun et al., 2026; Wang et al., 2026). Appendix Table 13 checks whether each work reports position-resolved commit/lock timing and varies block length. These works alter or analyze decoding schedules, but the cited evidence does not establish the same active-block recorder intervention; the scoped comparisons do not support a field-wide absence claim. Our supporting content analysis instead relates first-step confidence to later recorder lock ordering; it is not a new remasking policy or evidence for the five-arm mechanism.

Measurement studies likewise show that probes inherit task-format confounds, prompts can under-retrieve stored knowledge, and analysis choices shape the reported representation (Sahoo et al., 2026; Jiang et al., 2020; Belinkov & Glass, 2019). Construct validity therefore requires matching an instrument to its supported target (Westreich & Greenland, 2013); preregistration separates planned from outcome-contingent analysis but does not validate that target (Nosek et al., 2018). Our two-layer record applies this principle without the causal attribution of mediation analyses (Stolfo et al., 2023). Appendix Table 14 gives the observation-window and scale comparison, including the distinct monitoring example of Feng et al. (2024). The post-treatment amendment remains visible, the

min-max result is restricted to strictly- L -supported candidates, and the DFT control covers only the tested sinusoidal residual family.

Reproducibility statement. Reproducing the reported analyses requires the listed event streams, analysis scripts, and private model checkpoint. The current submission source package does not include them, so no numeric rebuild was run from that package.

6 CONCLUSION

Under the tested active-block recorder, the registered selector returns the configured block length across five arms. The matched arms, non-nested probes, bootstrap stability, and rank-sum comparison support that recorder-level description, but they do not provide null-calibrated detection or distinguish model-intrinsic state, sampler state, and recorder observability. The practical reporting recommendation from this case study is to state what is reachable from set composition before data and then separately state what measured magnitudes show. Position-decomposed measurements should be calibrated to the schedule that recorded them, without inferring a mechanism or cross-bed relation that the intervention did not measure.

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A PROTOCOL CHRONOLOGY, MEASUREMENT CANON, AND SENSITIVITIES

A.1 EXPLORATORY SPECTRAL RESIDUAL CHECK

For a nested candidate such as $\mathcal{B}_{32} \subset \mathcal{B}_{16}$, a *negative* $\text{sep}(32)$ is not evidence of absence: the extremum comparison cannot recover every possible residual shape. An *exploratory* (post-data, not pre-registered) check therefore asks whether a *non-nesting* estimator recovers a residual source period that sep misses. We use a DFT periodogram: for each candidate P , $G(P) =$

$\|\sum_r d(r)e^{-2\pi ir/P}\|/\|d\|$ is the normalized spectral magnitude of the adjacent-difference profile at frequency $1/P$, with a permutation null (shuffle $d(r)$ positions; 4,000 replicates) and family-wise q_{95} over all candidates. Harmonic testing follows the periodogram tradition (Fisher, 1929); our finite-grid threshold is a bespoke single-step maximum over candidates, using the standard permutation max-statistic construction (Nichols & Holmes, 2002). We also run a direct two-sample test of the two $L=16$ -boundary subsets in Table 6 (the odd-16 chain vs. the would-be- $\mathcal{B}_{32}$ positions), a reachable test of residual 32-amplitude modulation.

Case	$G(8)$	$G(16)$	$G(24)$	$G(32)$	$G(48)$	$G(64)$	$q_{95}@32$
Synthetic, residual-32							
present	3.58	3.58	—	4.86	—	—	1.74
Synthetic, no residual	4.01	4.01	—	0.25	—	—	1.75
Real L16 measured	3.79	3.70	0.15	0.23	0.36	0.18	2.15 (FW)

Supporting check	Statistic	Threshold / reference	Reading
Injected residual	Residual-32 amplitude 0.5 over comb-16 amplitude 3.0; $G(32)=4.86$	$q_{95}=1.74$	fires
No-residual synthetic	$G(32)=0.25$	$q_{95}=1.75$	null
Real L16 remaining	$G(24)=0.15$, $G(48)=0.36$, $G(64)=0.18$	per-candidate $q_{95} \approx 1.71\text{--}1.78$	no residual candidate
Comb harmonic	$G(8)/G(16)=1.000$ (pure comb), 1.024 (real L16)	1.0 reference ratio	recorder-comb harmonic
Power sweep	Amplitude 0.15: no fire; 0.20: fire; grid step 0.05 (5.0%–6.7% of comb amplitude 3.0)	q_{95} above	transition bracketed
Odd-16 vs. would-be- $\mathcal{B}_{32}$	Mean $ \Delta d =0.036$; permutation $p=1.0$	$\alpha = 0.05$	no amplitude split

Table 7: Spectral (DFT) periodogram of the adjacent-difference profile $d(r)$ — an *exploratory*, non-nesting check (Section A.1). $G(P)$ is the normalized spectral magnitude at frequency $1/P$; “fires” means the 32-component exceeds its permutation q_{95} (4,000 replicates). The upper panel completes the synthetic diagnostic grid and reports every real candidate; the lower panel separates the supporting power, harmonic, and two-sample checks from their interpretation. The synthetic residual fires, whereas the comb-only synthetic and real L16 profile are null at the target period. On real L16, no candidate in the tested grid is significant; only the recorder-comb harmonics fire. The power sweep brackets sensitivity for the tested sinusoidal family only, and the direct boundary-subset comparison finds no split. Provenance is in Table 9.

Development control. On the recorder-consistent synthetic grid, with a clean sinusoidal residual-32 component overlaid on the $L=16$ comb, sep returns $P^*=16$ (blind) while the spectral 32-component fires; on the no-residual synthetic comb, both are silent (Table 7). This is a positive control for the tested sinusoidal family, not validation against arbitrary residual shapes.

Real L16 arm. The tested candidate grid does not exceed its family-wise threshold at $P = 32$, and the boundary-subset test finds no amplitude split. The registered power sweep brackets the tested-family transition and records its amplitude range in Table 7. Hence the measured arm does not trigger for this tested sinusoidal family and amplitude range; arbitrary residual shapes are not ruled out (Table 9).

A.2 REGISTRATION CHRONOLOGY

Timestamp	Status	Recorded event
2026-08-20 17:55	Pre-data	Original registration filed before any GPU submission.
2026-08-20 18:09	Pre-data	Amendment v2 replaced the failed signed mean-difference score with the min-max estimator and registered the selection-and-stability criterion.
2026-08-20 23:07	Treatment	Both treatment arms ran (0.73 GPU-h).
2026-08-21 17:36	Post-data	Amendment v3 withdrew the model-intrinsic branch.
2026-08-21	Pre-data (probe)	Non-nested canvas-96 probe registration and amendments preceded its own run (0.38 GPU-h).

Table 8: Registration and amendment chronology. The estimator and selection criterion were fixed before treatment; the model-intrinsic decision branch was withdrawn after treatment data were observed, so the chronology documents a scope change rather than a pre-treatment mechanism verdict.

A.3 ARTIFACT INDEX

Table 9: Claim-to-artifact map. This is the sole table that prints workspace-relative artifact paths; all narrative references point here. At the analysis freeze, full provenance continued in `evidence/claim_evidence_map.tsv`, `evidence/method_provenance.tsv`, and `evidence/experiment_registry.json`; these ledgers are not included in the current submission source package.

Claim / role	Workspace-relative artifact	SHA-16
L32 development control	results/blocklen_probe/G0_VERDICT_r3.json	984b4ff92fe462ad
L16/L64 period	results/r29_l16l64_sawtooth.json	d462dff346604321
Cross-arm bootstrap, rank-sum, stability bands	results/r295_joint_bootstrap_extremum_robust.json	9a40c56c7c7a8243
Within-block selectivity gate and size audit	results/decomposition_control_r314/r316_sep_withinblock_permnull.json; results/decomposition_control_r314/r320_margin_selectivity.json	f623432c139e440c; e7d1c3115253589e
Boundary/interior cardinalities	results/decomposition_control_r314/r316_bcardinality_per_arm.json	e3a1c7b2b3c6915a
Content statistic and interval	results/r10_poslevel_gates/r10_poslevel_gates_v2.json	0af4f3c03e6c6f6d
Content null ladder	results/r274_bin1632_null_ext.json	b5d91fdc0b827018
Per-permutation nuisance refit	results/r259_beta_recompute/r259_beta_recompute.json	d3b27b3c87c924ce
Bit-exact nuisance-refit replay	results/r259_replay_B/r259_replay_B.json	85b875d858d0db19
One-flip three-bucket classification	results/r311_oneflip/r311_oneflip.json	47735637e3777a3d
Position-fake control	results/r14_margin_sweep/r14_margin_sweep.json	91da8f8b91a6b6b9
L16 measured nested-candidate check	results/r58_z1b_L16_truepair.json	10530a1b478ff492
L64 mid-block comparison	results/r58_z2b_l64_neg_evidence.json	470908f1400e7a1c
Reachability table	results/r147_reachability_table/r147_reachability_table.json	52a983b1c2fade9e
Strict-support synthetic diagnostic	results/r147_entailment_demo/r147_entailment_demo.json	ee4f1ad85d5492a2
Spectral family and power grid	results/r259_non_nesting_spectral/r259_non_nesting_spectral.json; results/r267_spectral_power_table.json	alb52a418ace4e07; 33768616bfa41ecc
Rank-transform sensitivity	results/r265_rank_transform_invariance.json; results/r299_orthogonal_separation_r299.json	f1760ce13789a67d; d90a4d56635f3771
Figure 2 manifest	results/r265_fig_dprofile_manifest.json	40079e3c19996a01
Control subset and row checks	results/r269_l32_control_subset_checks.json	e5455d953fad8367
Cross-environment reproduction	results/r269_estimator_byte_and_arm_recheck.json	2c5f824c0ef215db
Image-generation audit	build/image_generation_record.json	c019dbf2309f460b

A.4 MEASUREMENT CANON

For blockwise decoding observed through an active-block recorder, a position-resolved analysis should:

1. record block length, per-block steps, and generated-canvas coordinates;
2. calibrate within-block windows to the generating run rather than a fixed model constant;
3. report reachability twice: *a priori*, what estimator algebra and boundary-set composition imply; *from events*, what the measured magnitude ordering implies. In particular, $\mathcal{B}_{32} \subset \mathcal{B}_{16}$ does not force $\text{sep}(32) \leq 0$;
4. perturb the schedule when attributing a position-resolved diagnostic;
5. test content claims against a structure-preserving null and report its granularity; and
6. mark cross-bed relationships unmeasured until the corresponding intervention is run.

This is a measurement canon, not a mechanism claim. The registered estimator returns $P^* = L$ on the displayed *strictly L-supported* profiles: non-degenerate jumps occur on all L boundaries and no L -multiple's exclusive-boundary residual outscore the L -boundary minimum. It is not a guarantee for arbitrary residual profiles, and identifying commit internals would require instrumentation of sampler-state transfer.

A.5 SELECTIVITY DISCLOSURE (PERMUTATION-GATE SIZE AUDIT)

Original-gate size. The original within-block threshold was anti-conservative. Across the five arms, $10/25 = 40\%$ of wrong candidates (non-true- L grid points) were flagged by the original gate, so the nominal family-wise error of at most 5% was not realized. Table 10 gives the per-arm audit. The cause is that edge-including within-block permutation destroys the non-target global level/trend structure of $d(r)$, shifting the null downward and loosening the threshold.

Corrected margin and cross-condition red-team. Table 5 replaces that level gate with the top-minus-second separation margin and a discrete exact p . As a known-clean red-team, globally (cross-block) shuffling the same $d(r)$ constructs a structureless reference. The corresponding probabilities are consolidated in Table 10: four of five cross-condition checks are below 5% and L24 is again marginal. This pattern is consistent with schedule-aligned structure rather than within-profile jitter, but it does not calibrate the full selector. A genuinely clean arm has no blocks to permute, however, so this red-team is a cross-condition comparison, not a per-arm data-shuffle false-positive rate.

Arm	Original-gate wrong candidates	Cross-condition p_{margin}
L16	4/5	5.0×10^{-4}
L64	3/5	2.0×10^{-3}
L24	0/5	5.5×10^{-2}
L48	0/5	1.8×10^{-2}
L32 control	3/5	1.5×10^{-3}
Aggregate	10/25 (40%)	4/5 below 5%

Table 10: Selectivity-gate audit by arm. The original threshold flags 10/25 wrong candidates, whereas the cross-condition margin red-team is below 5% in four of five arms; L24 is marginal. These cross-condition probabilities are not per-arm data-shuffle false-positive rates.

Reporting rule. Any threshold-type permutation gate used here for selectivity reports three quantities together: the wrong-candidate hit rate, a known-clean-arm or explicit cross-condition check, and the discrete exact p . The 10/25 audit and the margin correction therefore accompany every selectivity claim based on this within-block gate.

A.6 THREE-PART EVIDENCE DECOMPOSITION

For auditability, we separate the recorder-relative object $d(r)$, the pre-treatment min-max decision procedure, and the scoped empirical finding that P^* follows L on the tested recorder beds. The shared-cluster bootstrap and rank-sum check support the decision procedure; the non-nested canvas-96 probes bound the empirical finding. Table 9 maps each component to its verifier. This decomposition is a falsification aid, not three independent novelty claims or a mechanism attribution.

A.7 SUPPORTING CONTENT ASSOCIATION ON A SEPARATE BED

Estimand τ . On the separate L32/NFE32 bed, one unit is a decode-block with resolved positions $\mathcal{R}$; c_r is initial confidence and $o_r = r \bmod 32$ is block offset. The estimand is the median per-unit rank association after residualizing confidence on offset:

$$r_u = \frac{\sum_{p < q \in \mathcal{R}: \ell_p \neq \ell_q} \text{sgn}(\ell_q - \ell_p) \text{sgn}(\tilde{c}_p - \tilde{c}_q)}{\#\{p < q \in \mathcal{R} : \ell_p \neq \ell_q\}}, \quad \tilde{c}_r = \text{rank}(\text{rank}(c_r) - \hat{\beta} \text{rank}(o_r)), \quad (3)$$

Here the registered statistic fits $\hat{\beta}$ once on the observed data and shuffles the resulting $\tilde{c}$ within offset-rank strata. A sensitivity check instead refits $\hat{\beta}$ inside every permutation replicate. Earlier lock with higher residualized confidence contributes +1. Inference is rooted at decode clusters: the 500-replicate interval resamples decodes and carries their block units with multiplicity. The position-fake keeps the observed amplitude but reassigns it to positions, destroying the observed content-position pairing. This differently budgeted bed supplies supporting, not cross-block-length, evidence.

Quantity	Estimate / band	Null q_{95}	Null readout
Content-layer τ_{res}	0.2722 [0.2649, 0.2812]	BIN16 0.20235	0/2000 exceed
Refit- $\hat{\beta}$ sensitivity	0.2722	BIN8 0.13545	0/2000 exceed
Same-amplitude position fake	max. margin +0.00979	observed margin +0.12837	below observed

Table 11: Content-versus-structure result on the separate $L = 32$, NFE-32 bed. The statistic τ is the median decode-block rank association between earlier lock and higher confidence after offset residualization. BIN16 is the load-bearing conservative, non-degenerate null; the per-replicate nuisance refit is a separate BIN8 sensitivity and replays bit-exactly. Table 12 reports the full granularity ladder and three-bucket one-flip classification. This supporting analysis is not a cross-block-length experiment.

The full-granularity ladder in Table 12 is essential. The observed association clears every non-degenerate restricted null through BIN16, but matches the zero-power point mass at BIN32. Thus content association is established at moderate granularity, never at the finest registered granularity; the construction-specific BIN32 line is not interpreted as a lower bound. Its behavior under $L \in \{16, 64\}$ is unmeasured, and the two analyses are not claimed to be mutually calibrated.

A.8 CONTENT ACCOUNTING AND NULL GRANULARITY

Accounting / sensitivity check	Recorded value	Interpretation
Content-resolved block-units	6,178 of 6,274 candidates	Terminal-resolved units excluded
Resolved comparable pairs	1,571,854	Both positions resolve
Within-unit support	Median 4 distinct locks; median resolved-pair fraction 0.575	Unit-level order is observed
Resampling	n_boot=500; R_null=2000	Decode-cluster interval and restricted null
K-ladder restricted-null q_{95}	$K=1$: 0.00486; BIN4 0.09276; BIN8 0.14384; BIN16 0.20235; BIN32 0.27221 (point mass)	With $\tau_{\text{res}}=0.2722$, margins at $K=1$, BIN4, BIN8, and BIN16 are 0.26735; 0.17945; 0.12837; 0.06986; BIN32 is zero-power by construction and carries no lower-bound reading
Nuisance- $\hat{\beta}$ sensitivity	Refit inside each BIN8 replicate: $q_{95}=0.13545$; fixed-fit BIN8 $q_{95}=0.14384$	0/2000 exceedances; bit-exact replay. BIN16 remains the load-bearing conservative non-degenerate arm
One-flip sensitivity ($1/n_{\text{selected}}$)	$n_{\text{selected}}=6178$; all 6178 residual-sign flips give $\Delta=0$	DEGENERATE, by construction; no pass claim. The null-specification ladder, not this identity arm, carries the sensitivity conclusion

Table 12: Unit accounting and restricted-null sensitivity for Table 11. A block-unit is rooted in its decode cluster; decodes are resampled with multiplicity. The registered line fits the nuisance slope once; the displayed sensitivity refits it inside each replicate. One-flip granularity is $1/n_{\text{selected}}$, not permutation spacing $1/R$. Under the three-bucket contract, maximum single-point effect $\phi = 0$ is DEGENERATE (by construction) and carries no pass/fail claim; $0 < \phi \leq \text{margin}$ would pass and $\phi > \text{margin}$ would fail. BIN32 is likewise zero-power by construction. The full sweep therefore limits the positive finding to moderate granularity. Provenance is in Table 9.

A.9 CLOSEST-WORK SETUP COMPARISON

Table 13: Row-wise setup comparison for the cited nearest work. Entries state the object and observation window, the schedule/measurement action, and the resulting contrast with this recorder audit, including whether each reports *both* layers of our two-layer rule (a-priori reachability *and* from-events magnitude). These scoped contrasts do not assert that an entire research area lacks schedule diagnostics or mechanism analysis.

Work	Object and observation window	Schedule action / measurement	Measurement-layer contrast used here
Austin et al. (2021)	Structured discrete-state denoising; model/sampler trajectory	Schedule belongs to the formulation	Background; reports neither a-priori reachability nor from-events magnitude
Zheng et al. (2024)	Reparameterized text diffusion; generated trajectory	Reparameterizes a fixed generative process	Changes model parameterization; we hold the model fixed and perturb the recorder-facing partition
Lou et al. (2024)	Distribution-ratio estimation for discrete diffusion; likelihood/generation	Schedule supports the generative objective	Targets modeling/likelihood, not boundary-set reachability
Shi et al. (2024)	Simplified masked diffusion; model/sampler output	Simplifies the diffusion formulation	Supplies the masked-diffusion setup, not the two-layer recorder audit
Li et al. (2025)	Variable-length inference; whole-decode outcomes across lengths	Changes denoising length	Schedule change used for decoding; ours fixes total NFE over a block-partition co-movement
Sun et al. (2026)	Bayesian dynamically structured decoding; evolving decode	Adapts decoding structure	Schedule is the method’s action; ours treats it as a diagnostic perturbation

Continued on next page

Table 13 continued

Work	Object and observation window	Schedule action / measurement	Measurement-layer contrast used here
Wang et al. (2026)	Reversible diffusion decoding; revisable trajectory	Permits decisions to be revisited	Reversibility changes the trajectory; ours audits masked active-block events
Sahoo et al. (2026)	Hidden-state linear probe; language-model input/activations	Deconfounds measurement from format	Warns instrumentation confounds can induce a readout; reports measured magnitudes, not a-priori set composition
Jiang et al. (2020)	Prompt query of LM knowledge; output distribution	Optimizes prompts	Frames any measurement as a lower bound; distinguishes reachability from measured magnitude for facts
Belinkov & Glass (2019)	Analysis/probing methods for representations; activations	Chooses analysis method	Choice of method changes what is measured; no recorder-window partition perturbation
Feng et al. (2024)	Propositional probes of latent world states; activations	Monitors via probes	Shows recorded layer can diverge from behavior; does not perturb the recording window
Stolfo et al. (2023)	Causal-mediation attribution; internal states	Intervenes on internal states	Attributes mechanism; we do <i>not</i> and report only recorder-level reachability
This work	Identity-lock adjacent difference; active-block event window	Matched block-length intervention plus required allocation	Separates a-priori set composition from event-derived magnitude; does not claim mechanism

Table 14: Nearest-neighbor positioning against the closest conceptual measurement works. Each prior operates on a different measured dimension; the table states that relation and the checks already run on the recorder-window period-separation dimension. Artifact paths and SHA-16 prefixes are in Table 9; this is not a mechanism attribution.

Closest conceptual work	Their measured dimension (claim)	Same dim. here?	Testable gap this recorder audit runs (already run)
Jiang et al. (2020)	Prompt-level QA recall/accuracy on factual knowledge; measurement as a lower bound on stored facts	No — conceptual source of our two-layer split; their scale is knowledge retrieval	Differential separator, non-nested canvas-96 probes, and five-arm rank-transform invariance (Table 9)
Sahoo et al. (2026)	Hidden-state probe AUROC/accuracy; confound driven by task format, collapsing after deconfounding	No — deconfounds task format, not the recorder partition	Matched composite partition intervention plus a non-nested canvas-96 instrumentation self-check (Table 9)
Feng et al. (2024)	Propositional-probe AUROC for latent world states; recorded layer vs behavior divergence	No — compares layer to behavior, does not perturb the recording window	Fixed model/NFE recorder-window perturbation, paired bootstrap, and rank-sum separation (Table 9)

B REPRODUCIBILITY AND ARTIFACT MAP

Execution cost, shard completion, and event-row validation are reported in Table 15, separately from provenance, so that spend is never conflated with a completeness check. Table 9 records the workspace-relative provenance paths and whole-file SHA-256 prefixes used at the analysis freeze.

The current submission source package contains the manuscript, bibliography, and two figures, but not the listed result, script, and evidence bundle; the table is therefore a provenance index rather than a self-contained artifact release.

Determinism. Decoding uses temperature 0 with *remasking=low_confidence*; under the fixed decoding rule and fixed NFE schedule, token selection is deterministic on a fixed canvas, so the identity-lock profile $m(r)$ and its adjacent difference $d(r)$ are deterministic functions of (model, checkpoint, seed, question set, NFE schedule). Min-max separation $\text{sep}(P)$, period selection, and every reported spectral value are deterministic properties of these recorded profiles (only the bootstrap / permutation-replicate quantities carry sampling randomness). Determinism is scoped per (checkpoint, seed, question assignment, NFE schedule); cross-seed variance is not the pre-registered estimand.

Environment. The registered analyses ran under Python 3.11.13, NumPy 2.4.6, pandas 2.3.1, PyArrow 21.0.0, PyTorch 2.7.1; the model is a locally registered LLaDA-8B-Instruct-scale masked dLLM, blockwise decoded over GSM8K (Cobbe et al. 2021), a fixed 300-question canvas at 256 positions and 48 total NFE per arm. The checkpoint is a private local artifact (registry per the project evaluation ledger) and is *not* byte-identical to the public LLaDA-8B-Instruct release; we verify its magnitude is at the same scale but do not assert release equivalence. The analysis-time manifest and core-script paths are indexed in Table 9, but those files are not included in the current submission source package.

Cross-environment reproduction. The estimator source was touched after the submission manifest recorded its hash, so rather than assert equivalence we re-derived every arm from raw events with the *current* bytes in a second software environment (Python 3.12.3, NumPy 2.5.2, pandas 3.0.5). All three arms reproduce their sealed sep vectors and selected periods *exactly*, so the second-environment Table 3 values equal the first-environment values bit-for-bit and are not restated as a duplicate column. The complete arm-by-candidate values and both estimator hashes are indexed in Table 9. This reported replay cannot be independently rerun from the current submission source package alone.

Reproduction boundary. Decode uses *seed=20260820* and the frozen question subset $\text{decode_id} \in [345, 645]$. The event schema, estimator paths, and analysis-time digests are recorded in Table 9, but the current submission source does not ship the event streams, executable scripts, or private checkpoint and hence does not support a numeric rebuild. The checkpoint cannot be re-released without authorization; an anonymous supplement of releasable events, scripts, and hashes remains subject to provenance approval.

Execution accounting	Recorded value
Hardware	eight GPUs
Timing	330 running s = 0.73 GPU-h; 130 queueing s reported separately and excluded
Shard completion	$L = 16$: 4/4; $L = 64$: 4/4
Event-row check	$L = 16$: 153,600 = 300×512 ; $L = 64$: 499,200 = 300×1664 ; $L = 32$ control: 268,800 = 300×896 rows for $\text{decode_id} \in [345, 645]$ of the frozen NFE-48 pool

Table 15: Execution cost and completion checks for the two new treatment arms and the frozen control subset. Cost is reported for the GPU submission that produced the $L=16$ and $L=64$ arms; shard and event-row checks establish completion, whereas the control arm reuses a pre-existing pool.

AI-assisted preparation disclosure. Image generation produced qualitative Figure 1; the selected candidate passed an element-level semantic audit and carries no measurements. Figure 2 remains data-derived. An AI assistant edited prose and LaTeX; all claims, numbers, citations, and hashes were checked against the listed evidence.

C NOTATION AND TERMINOLOGY

Table 16 collects the estimator notation and verdict vocabulary; Table 17 collects the bed names, the treatment variable, and the content-layer terms. Each entry names the section that defines it.

Table 16: Estimator notation and legacy decision-table vocabulary; terms point to their operative definition and do not attribute mechanism.

Term	Meaning (defining location)
$\ell_{i,r}$	Identity-lock time: earliest recorded step after which decode i 's token at relative position r never changes again (Eq. (1)).
$m(r), d(r)$	Per-position median lock time over decodes, and its adjacent difference $m(r+1) - m(r)$ (Section 2).
$\mathcal{B}_P, \mathcal{I}_P$	Candidate- P boundary set $\{r : r \bmod P = P - 1\}$ and its complement, the candidate interior (Section 2).
$\text{sep}(P), P^*$	Min-max period separation $\min_{\mathcal{B}_P} d - \max_{\mathcal{I}_P} d$, and its maximizing candidate (Eq. (2)).
SELECTED	$\text{sep}(P^*) > 0$, bootstrap modal period $= P^*$, mode fraction ≥ 0.95 (registered operational criterion, not a null-calibrated test; Section 2).
REFUTED / NOT-SUPPORTED	Measured $\text{sep}(P) < 0$ for a candidate that the decision table treated as refutable / for the originally registered intrinsic-32 conjunct, whose branch was withdrawn post-data (Table 6).
A-PRIORI layer	What follows from the estimator's algebra alone, before data — e.g. boundary-set nesting (Section 2).
FROM-EVENTS layer	What follows only once measured magnitudes exist — e.g. the sign of $\text{sep}(32)$ (Section 2).
Strictly L -supported	Recorded profile with non-degenerate jumps on all L -block boundaries such that no L -multiple's exclusive-boundary residual outcores L 's boundary minimum (Section 3.1).
$G(P)$, perm. q_{95} , fam. q_{95}	Normalized DFT magnitude of $d(r)$ at frequency $1/P$; its per-candidate permutation 95th percentile; the family-wise version over all candidates (Section A.1).

Term	Meaning (defining location)
L16 arm, L64 arm	The two new matched treatment arms at block length 16 and 64, NFE 48 (Table 1).
L32/NFE48 control	Frozen matched positive control at block length 32, NFE 48: the registered $\text{decode_id} \in [345, 645]$ subset, $n=300$ (Sections 2 and 3.1).
L32/NFE32 content bed	Separate, differently budgeted bed used only for the content-association diagnostic (Section A.7).
Partition-plus-required allocation	The composite treatment: changing the block partition necessarily changes the per-block step allocation at fixed total NFE (Table 1).
Measurement-class label	PERIOD-FOLLOWS-BLOCK-LENGTH: a post-data label stating that the selected boundary period follows block length on the tested bed under the active-block recording window; not a mechanism attribution (Sections 1 and 3.1).
BIN8	Restricted null that permutes residualized confidence within 8 equally populated offset-rank bins, preserving offset structure to 4-position granularity (Sections 2 and A.7).
conf_0	Position- r initial confidence at block-commit: the model's confidence for r at the first recorded step of its block (Section 2).
Block-unit; resolved position	One decode-block of the content bed; a position whose lock time is defined and that never flips after lock. Terminal-resolved units, which present no within-block pairwise order, are excluded (Table 11).
τ, τ_{res}	Median over units of the per-unit rank association between lock time and confidence; the offset-residualized (content-layer) version (Eq. (3)).

Table 17: Bed, treatment, and content-layer terms; the matched setting and separate content bed remain distinct.